University of CINCINNATI

Syllabus

Fall 2025, MATH1045, Applied Calculus II, Section 001

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Office Hours: Tuesday, 9am - 12pm (in the MASS Center – FH2133)

Students are expected to review and follow the linked course policies, university policies, and community resources on the following topics in the UC Student Resources Canvas site.

- Attendance
- Class Cancellation
- Academic Integrity
- Accessibility
- Title IX
- Counseling Services

Course Coordination and Communication

MATH1045 is a large enrollment course. To make sure you have an experience that is comparable to students in other sections, a course coordinator will make sure course policies are followed and that grading is fair across all sections. Our course coordinator is Dr. Inne Singgih (inne.singgih@uc.edu).

There are two Canvas for this course: **our section (001) Canvas** and the **All-Sections Canvas**. Our sections Canvas is where you can find materials, assignments, and announcements for our sections. The All-Sections Canvas is for common assignments and announcements across all MATH1045 sections.

The instructor reserves the right to make changes to the syllabus, including assignment due dates when necessary to meet learning objectives or when unforeseen circumstances occur. Any changes to our schedule, syllabus or policies will be communicated through **Canvas** as early as possible so students can adjust their schedules. It is your responsibility to make sure Canvas has a valid email address and it is recommended that you check this email daily. All course documents will be available on Canvas as well.

Course Overview, Description, Purpose

This is the second part of a two-semester sequence (MATH1044 and 1045) on calculus appropriate for students in business and life sciences. Topics covered include anti-differentiation, the fundamental theorem of calculus, functions of two variables, partial derivatives, maxima and minima, Lagrange multipliers and applications to probability and other areas.

First and foremost, this course will strengthen your mathematical THINKING and problem-solving skills. We'll achieve this by introducing you to new mathematical CONCEPTS, figuring out how to solve related problems, and using the concepts to interpret information. The main concepts we will work on are integration/antiderivatives and calculus with two variables.

This semester, we will cover 20 sections across 4 chapters:

- Chapter 5: Section 5.1 to Section 5.9
- Chapter 6: Section 6.1 to Section 6.6
- Chapter 7: Section 7.1 to Section 7.3
- Chapter 8: Section 8.1 to Section 8.2

Course Learning Outcomes

University of Cincinnati e-Curriculum Outcomes:

1. Numerically estimate the value of definite integrals using Riemann Sums and the area interpretation of the integral.
2. Use integration formulas and substitution to compute indefinite integrals for polynomials, rational functions, exponential functions, logarithmic functions and trigonometric functions.
3. Estimate the numerical value of definite integrals using technology.
4. Use the fundamental theorem of calculus to evaluate definite integrals, solve and interpret solutions of applied problems.
5. Calculate the area under a curve and the area of the region between two curves.
6. Apply integration techniques to solve and interpret solutions of applied problems such as present and future value of an investment, consumer's surplus, producer's surplus, elasticity, improper integrals and average value of a function on an interval.
7. Use probability density functions to compute and interpret probabilities as well as compute the mean and median of a distribution.
8. Compute partial derivatives of a function of two variables and use them to determine extreme values.

Ohio Transfer 36 TMM013 Business Calculus Learning Outcomes:
3.08, 4.01, 4.02, 4.04, 4.05, 4.06, 4.07, 4.08, 4.09, 4.10, 5.02, 5.03, 5.04, 5.05

Pre-requisites

To take this course you must: Have taken the following courses:
MATH1044 min grade C-, or MATH1060 min grade C-, or MATH1061 min grade C-.

Course Format

This class is an **in-person** class. In-person class meetings will NOT be recorded. Students at the University of Cincinnati are expected to attend class to meet the course's learning objectives. You are responsible for what we discuss in class whether you attend or not.

We are using **flipped content** delivery. Math is learned best by doing, which is why we will emphasize *doing* problems in class vs. a more lecture-based class. It is vital that you familiarize yourself with concepts by watching the lecture videos **BEFORE** class. The same lecture videos are provided across MATH 1045 sections. Students are to watch them prior to coming to class. In class we will emphasize discussing and working on various problems.

Course Materials

Optional Text: **Calculus Concepts**, 5th edition, by LaTorre, et al.
The textbook is optional -- you can opt out from Cengage Follett Access (otherwise, the textbook price, despite reduced, will be added to your tuition). Opting out typically due the 15th day of the semester. More info is on the Office of the Bursar – Special Course Fees page.

You are **required** to have a physical graphing calculator. If you need to purchase one, please get a TI-84.
If you already have a TI-83 or TI-nSpire, those are permitted.

Phone apps, online graphing tools, or any AI-equipped devices are NOT permitted during tests.

You are recommended to bring a laptop/device with internet capability during class session.
If this is not possible, please plan to work with a partner who will bring one.

Assessments/Activities and Grading Policy

Your grade for this course will be assigned based on the following points:

Block Final Exam (cumulative)	20%
Chapter 5 Test	17%
Chapter 6 Test	17%
Chapter 7 and 8 Test	17%
Instructor Points (10% PQs, 7% PCQs)	17%
Projects (6% each)	12%

Each of these categories is explained on the next page of this syllabus.
The last day to withdraw (no refund) is November 21.
Any changes on this deadline and refund deadline information can be found on the Office of the Registrar page.

Letter Grades

Letter grades will be assigned based on the percentage of the weighted total points received. We expect to use the typical ten-point grading scale, with rounding at .5% as follows:

Letter Grade	A	A-	B+	B	B-	C+	C	C-
Weighted Total Min	92.5%	89.5%	86.5%	82.5%	79.5%	76.5%	72.5%	69.5%

There is no curving in this course, but borderline cases (within 1% from grade cutoff) might be considered for a rounding up. The decision of the rounding will be based on the discussion between the section instructor and the course coordinator.

NP Grading Scheme

Students who complete the course without earning at least a C- will be assigned the grade of NP (Not Proficient). An NP does not lower your GPA; however, it also does not count as a passing grade, earn course credit or meet any pre-requisites. If this course is an essential part of your academic plan and you earn an NP as your final grade, you must retake the course successfully in a future semester.

Students who receive an NP might have earned enough points for a D+, D, or D- grade. These students can request a change in their grade from NP for up to one year after the NP grade was assigned. It is important to consider that D grades will affect your GPA and might affect your academic standing. Contact your academic advisor to decide whether to request this change or if you have questions about the effect of the NP in your academic program.

IMPORTANT: Student must academically attend class to be eligible for an NP; this is especially important for students receiving financial aid. Attending just one or two class meetings does qualify as academic attendance.

More information about academic attendance can be found on Office of the Registrar Academic Attendance page.

Explanation of Grading Components

Block Final Exam (cumulative) The final exam **will not** be on the day, time, and location listed on the university exam schedule, as ours is a **block exam**, scheduled by the Registrar's Office. Schedule TBA a.s.a.p. ***Do not make travel plans during final week!***	in-person on paper closed notes work individually no re-attempts Chapters 5, 6, 7, 8, excluding 6.1
Chapter 5 Test Chapter 5 Test takes place during scheduled class time. Original date: Thursday 9/25 Friday 9/26 Retake date: Monday 10/6 or Tuesday 10/7	in-person on paper closed notes work individually 1 re-attempt Chapter 5
Chapter 6 Test Chapter 6 Test takes place during scheduled class time. Original date: Thursday 10/23 or Friday 10/24 Retake date: Monday 11/3 or Tuesday 11/4	in-person on paper closed notes work individually 1 re-attempt Chapter 6
Chapter 7 and 8 Test Chapter 7 and 8 Test takes place during scheduled class time. Original date: Thursday 11/20 or Friday 11/21 Retake date: Thursday 12/4 or Friday 12/5	in-person on paper closed notes work individually 1 re-attempt Chapter 7 and 8
Pre-Class Quizzes (PCQs) Lectures from each section will be delivered via Canvas videos. After watching each video, you'll be asked a few questions to apply what you've just learned. While PCQs are due before class, they are available until 12/5 and may be attempted until then.	online open notes work together receive assistance unlimited attempts **3** lowest dropped.
Participation Quizzes (PQs) In class, we will work on PQs. You are encouraged to work in a group, and the instructor will walk around answering questions. Each PQ is due at the end of the corresponding class day.	online open notes work together receive assistance 3 total attempts No extensions **3** lowest dropped.
Projects Projects guide you through applications of calculus using a variety of technology. **Please bring your computer and calculators to class on the dates for each project.** There are two (2) Projects this semester.	in-person on paper submit online open notes work together receive assistance 1 re-submission **0** None dropped

For Chapter Test, if you missed the original, then you only have the retake quota left (only have 1 attempt).

Assessment Policies

It is very important to us that the assessments in this class are fair, that grading is consistent, and that nobody receives unfair advantages. This is why we will strictly enforce the policies below:

- Participation in the common assessments (Chapter Tests and Final Exam) is mandatory.
- Cell phones and other communication devices are prohibited. They must be put away during all assessments.
- Students **must** present a photo ID and sign out with the instructor/proctor.
- Students with accommodation should forward the email from AR Office to their instructor as soon as possible. Registered students should schedule their assessments with Testing Services on the scheduled dates.
- If you have **documentation** of a conflict during any test (examples below), please submit the appropriate evidence to your instructor as soon as possible.
  - Hospitalization/Doctor's Visit (documentation excuses you on exam date)
  - Official University Business (travel letter for athletics, clubs, internships, etc.)
  - Conflict with other UC classes (submit your Catalyst schedule)
- If your documentation is approved, you will be given the opportunity to retake the assessment at a time scheduled by your instructor.
- If you believe there has been a grading error in an assessment, you can request the assessment to be re-graded within a week of your receiving it.

Final Exam

The Final Exam is scheduled by the Registrar's Office, **NOT** on the date listed on the Final Exam schedule. *Please do not make travel plans during exam week until you know the final exam date.* You may only request an alternate time for your Final Exam if you have a conflicting time with another course's final exam.

Academic Integrity

The University Rules, including the Student Code of Conduct and other documented policies of the department, college and university related to academic integrity including the assessment policies listed above will be enforced.

(AI) Artificial Intelligence/Apps/Extensions

You should **not** use any kind of AI tools (software/apps/extensions) to produce work for this class without doing any work yourself. While these might help with checking your solutions, if misused they often provide incorrect and incomplete information. You might trick yourself into believing your calculus ability is stronger than it really is or worse, confuse yourself by introducing advanced topics beyond the scope of this course. Besides, the majority of course credit comes from in-person assessments where these resources are strictly prohibited. Verified use of AI on the Projects, Tests, or Final Exam will be reported as Academic Misconduct. If you are in doubt as to whether you are using an online learning support appropriately, I encourage you to meet with me for advice.

Authentic Individual Work

Although some activities of the course involve collaboration, Quizzes, the Midterm Exam and the Final Exam must be completed on your own. Violating the student code of conduct, (e.g. cheating, aiding or abetting) on these assessments, will be reported as Academic Misconduct.

The sanction for Academic Misconduct will be a failure of the course.

Please also review the College of Arts and Science's Two-Strikes Policy. We take these violations seriously.

Section 001 Classroom Procedures/Policies

Attendance Policy

On a class day without assessments, attendance is encouraged, but not mandatory. If you miss class due to illness or an important obligation, you're more than welcome to let me know, but it is not required.

However, it is essential that you attend in-person all scheduled Quizzes, Projects, the Midterm Exam and the Final Exam. **There is no option to complete these components remotely.**

Technology/Cell Phone Policy

You are welcome to use a laptop or tablet in this class if it contributes to your learning (taking notes, reading materials, etc.). However, electronics can be a major distraction in class. Specifically, cell phones, texting, email, and other social media are a powerful form of distraction. Please turn your cell phones on silent mode. I will ask you to put away your device if it is distracting you or others. If becomes a repeated issue, I will ask you to leave class.

Accessibility Policy

UC implemented a centralized platform to oversee the accommodation process called UC Accommodate. Students with special needs should register with Accessibility Resources and familiarize themselves with UC Accommodate platform as soon as possible to arrange reasonable provisions to ensure an equitable opportunity to meet all the requirements of this course.

The University of Cincinnati is committed to providing all students with equal access to learning opportunities. Additional information can be found on the Accessibility Resources webpage.

Student Religious Accommodations

Ohio law and the University's Student Religious Accommodations for Courses Policy 1.3.7 permits a student, upon request, to be absent for reasons of faith or religious or spiritual belief system or participate in organized activities conducted under the auspices of a religious denomination, church, or other religious or spiritual organization and/or to receive alternative accommodations with regard to examinations and other course requirements due to an absence permitted for the above-described reasons. Not later than 14 days after the first day of instruction in the course, a student should provide the instructor with written notice of the specific dates for which the student requests alternative accommodations. For additional information about this policy, please contact the Executive Director of the Office of Equal Opportunity and Access at (513) 556-5503 or oeohelp@UCMAIL.UC.EDU.

Intellectual Property Statement

All original instructor-provided content for this course, which includes handouts, assignments, lectures, and video content is the intellectual property of the instructor(s). Students enrolled in the course this academic term may use the original instructor-provided content for their learning and completion of course requirements this term, but such content must not be distributed or monetized in any way.

Student Resources

Supplemental Review Sessions

All MATH1045 students are invited to sign up for an optional 12-week program of supplemental review sessions (SRS) through MATH1096 (301-302). The time commitment is 1 hour and 20 minutes per week, for 12 weeks. You will review essential prerequisite material, work on problems, present solutions, and discuss and correct solutions under the guidance of upperclassmen and graduate assistants through the Learning Assistance Center. If you choose to sign up for the SRS, then they become compulsory for you. More information about SRS can be found on the SRS page.

Drop-in Study Tables

The Math and Science Support (MASS) Center provides study-table tutoring for this course. More information about the study tables can be found on the MASS page.

Tutoring and Coaching

The Learning Assistance Center (LAC) offers one-on-one tutoring for this course. Students may schedule individual tutoring appointments to improve their understanding of course materials and develop effective study strategies. LAC appointments are available Mon-Thurs 9am-8pm and Fri 9am-5pm. Students may schedule appointments online. More information about tutoring can be found on the Peer Tutoring page.

The LAC also offers Academic Coaching. Academic Coaches are high achieving UC upperclassmen and graduate students who provide one-on-one support to encourage success-building practices and habits in students. Coaching is not course specific, but applicable to all majors and courses. If you feel that "test anxiety" affects your performance, consider talking to an academic coach. More information about coaching can be found on the Academic Coaching page.